

Day 3: DNS Server ConfigurationTheory

Task 1 :- Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

ANS :-

1.Install BIND9 (Linux) :

Run this command in your terminal:

```
sudo apt update && sudo apt install bind9 -y
```

2.Add domain to config :

Open the zone config file :

```
sudo nano /etc/bind/named.conf.local
```

```
zone "musicmania.in" {  
    type master;  
    file "/etc/bind/db.musicmania.in";  
};
```

3.Create the zone file :

Create the new domain database file:

```
sudo nano /etc/bind/db.musicmania.in
```

Paste this simple DNS record layout :

```

$TTL      604800
@         IN      SOA      ns1.musicmania.in. admin.musicmania.in. (
                                2026100401 ; Serial
                                604800     ; Refresh
                                86400      ; Retry
                                2419200    ; Expire
                                604800 )   ; Negative Cache TTL
;
@         IN      NS       ns1.musicmania.in.
@         IN      A        192.168.1.100
www       IN      A        192.168.1.100

```

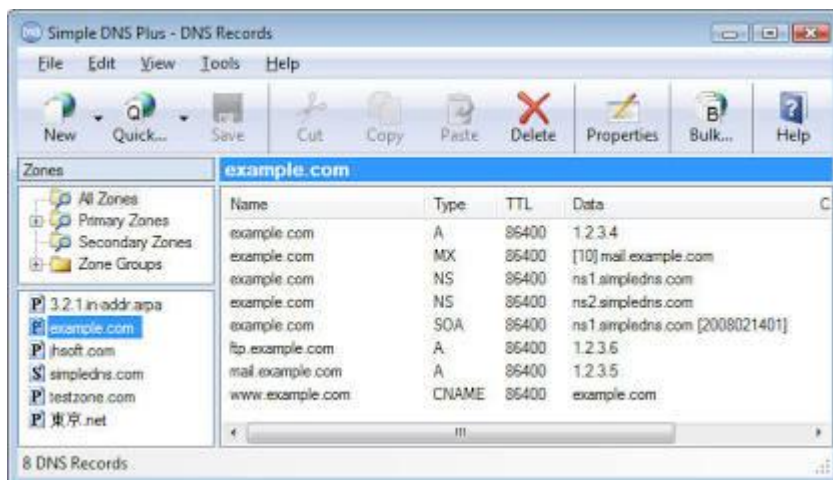
4.Restart and test :

Restart BIND :

```
sudo systemctl restart bind9
```

Test if it resolves :

```
dig @localhost musicmania.in
```



If you're on Windows using Simple DNS Plus:

1. Download and open Simple DNS Plus.
2. Click Records > Quick > Add New Zone.
3. Type musicmania.in and hit OK..

Task 2 :- Add an A record in your DNS server configuration to map `www.musicmania.in` to the IP address `192.168.1.50`, then use the `nslookup` or `dig` command to verify the mapping works.

ANS :-

1.Update the zone file :

Open your domain file :

```
sudo nano /etc/bind/db.musicmania.in
```

Find the line for `www` and change its IP to `192.168.1.50` :

```
www      IN      A       192.168.1.50
```

2.Restart DNS server :

Apply the changes :

```
sudo systemctl restart bind9
```

3.Verify with dig or nslookup :

Run either command to check :

```
nslookup www.musicmania.in 127.0.0.1
```

Look for `192.168.1.50` in the answer section.

Task 3 :- Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

ANS :-

1.Add CNAME record :

Open your zone file :

```
sudo nano /etc/bind/db.musicmania.in
```

Add this line at the bottom :

```
play      IN      CNAME    www.musicmania.in.
```

2.Restart DNS server :

Reload BIND :

```
sudo systemctl restart bind9
```

3.Test CNAME resolution :

Run this command :

```
nslookup play.musicmania.in 127.0.0.1
```

It should point play.musicmania.in to www.musicmania.in, which resolves to 192.168.1.50.

Task 4 :- Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.

ANS :-

1.Check config syntax :

Check config files for missing semicolons, dots, or syntax errors :

```
sudo named-checkconf  
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
```

Fix any errors shown on screen.

2.Check DNS service status :

Verify BIND is actually running :

```
sudo systemctl status bind9
```

If it's inactive or failed, restart it :

```
sudo systemctl restart bind9
```

3. Check system logs for errors :

View recent DNS system errors :

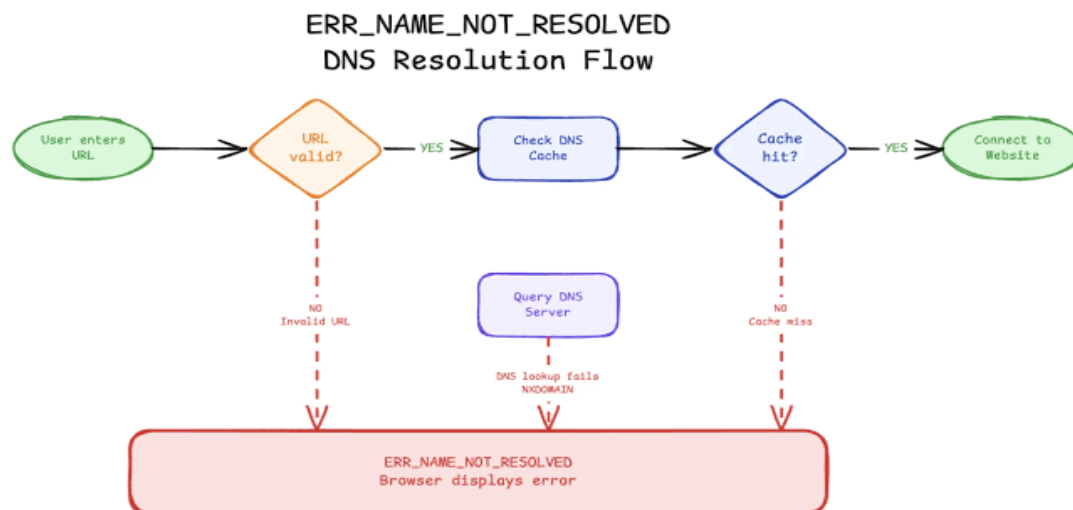
```
sudo journalctl -u bind9 -n 20 --no-pager
```

4. Test resolution directly :

Query local DNS server to check if it returns the IP :

```
nslookup musicmania.in 127.0.0.1
```

If it returns NOERROR and shows 192.168.1.50, your DNS is working properly.



Quick Solutions:

1. Check URL

- Verify spelling
- Check domain extension

2. Clear DNS Cache

- `ipconfig /flushdns`
- `sudo dscacheutil -flushcache`

3. Change DNS Servers

- 8.8.8.8 (Google)
- 1.1.1.1 (Cloudflare)

4. Debug with CLI

- `dig example.com`
- `nslookup example.com`